Learning Objectives

- To learn about Internal and External combustion engines.
- To learn about the engine parts, manufacturing procedure and the material used.
- To learn about the thermal energy produced in two stroke and four stroke engines.

5.0 INTRODUCTION

Thiruvalluar quotes "When water fails, functions of nature cease". In modern world "When automobile fails, functions of world cease". The mobility of the people and world will ceases when there is no automobile. Thus Automobile plays a vital role in transportation of people and goods from one place to another, even in between the continents.

The driving force required to drive the vehicle is provided by the Engine. Engine is considered to be the heart of an automobile. Based on Law of conservation of Energy, i.e., Energy can be transformed from one form to another form, engine is used to convert heat energy obtained by burning of fuel into mechanical energy and therefore they are called as Heat Engines.

5.1 PETROL ENGINE

In petrol engines, the heat energy is obtained by burning the petrol with air and this heat energy is converted into mechanical energy. Since petrol is also called as gasoline, this engine is also called as Gasoline engines.

The liquid state of petrol fuel is converted into vapour and it is mixed with atmospheric air. This air-fuel mixture is inducted into the engine, and it is burnt with the spark introduced by the spart plug. Hence, this engine is also called as spark ignition engine. This engine was invented by Nichloas Otto, German scientist in 1876. The engine is working based on otto cycle which is constant volume cycle. To burn a fuel, four operations is required namely, intake of fuel air mixture, compression of fuel air mixture, burning of furl air mixture and sent out the burnt air fuel mixture from the engine. Each operation is completed in each stroke of the piston and four stroke is required to complete a cycle. Hence it is called as four stroke engine.

5.2 DIESEL ENGINE

In diesel engines, the heat energy is obtained by burning the diesel with air and this heat energy is converted into mechanical energy. In 1897, Rudolph diesel invented the Diesel engine and hence the engine was called by his name "Diesel engine". This engine is working based on constant pressure cycle. In diesel engine, the air alone is intaken during suction stroke and it is compressed during the compression stroke. At the end of the compression stroke, diesel fuel is injected at high pressure which automoise the diesel fuel. The temperature of the compressed air is sufficient enough to start the combustion. Since air alone is compressed at high pressure, it will liberate more energy than petrol engine. Hence this engine is used on trucks, buses and heavy vehicles.

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